

Lazy Prices on the S&P 100: a replication and an overfitting audit

Ignacio Queipo de Llano*

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Abstract

Cohen, Malloy and Nguyen (2020) show that firms whose annual report changes little from one year to the next outperform firms whose report changes a lot, and that a long-short portfolio on this signal earned 30 to 60 basis points a month on the CRSP universe from 1995 to 2014. I replicate the strategy on the current S&P 100 with free data (1,792 10-K filings from EDGAR, yfinance prices, Fama-French factors) over March 2009 to September 2026. The long-short portfolio earns -0.92% a year with an FF5+MOM alpha of -0.93% ($t = -0.49$); the twelve-month event-time abnormal return is -1.8% ($t = -0.60$); the within-cohort rank correlation between text change and forward return is indistinguishable from zero for every measure. Searching over 54 variants produces a best annualised Sharpe of 0.54, but the deflated Sharpe ratio of that winner is 0.13 and its Sharpe sits at the 99.5th percentile of a placebo distribution obtained by shuffling scores within cohorts. On this universe and period there is no detectable effect. The note documents why that is the expected outcome given the sample and how the tools of Bailey and López de Prado keep the best of 54 backtests from being reported as a finding.

1 Introduction

Lazy Prices argues that annual reports are read by few and changed by fewer: when a firm rewrites large parts of its 10-K, the rewrite usually carries bad news, concentrated in the risk factors (Item 1A) and the management discussion (Item 7), and the market absorbs it slowly. The authors measure the change with four similarity measures between consecutive 10-Ks and 10-Qs, sort firms into quintiles, and find that a portfolio long “non-changers” and short “changers” earns 30 to 60 basis points a month of abnormal return over the following year, with the effect concentrated in the months after the filing.

This note asks a narrow question: does the signal show up in the largest and most scrutinised US companies, using only public, free data? The answer is no, and the interesting part of the exercise is establishing that the “no” is robust to the researcher’s own degrees of freedom. Section 2 describes the data and its limitations, Section 3 the construction, Section 4 the results, Section 5 the selection-bias audit and Section 6 what I would conclude.

2 Data

Universe. The 99 companies of the S&P 100 as of September 2026, identified by CIK. This is a *current* list, so every firm in it survived and grew into the index; firms that were members

*Code, data and this note: <https://github.com/iqueipopg/lazy-prices>. All numbers in the text are macros generated from the result files by `python -m lazyprices.report`.

in 2010 and later shrank, were acquired or failed are absent. Five companies changed CIK after a holding-company reorganisation (Exxon Mobil, Alphabet, Disney, BlackRock, Medtronic); the predecessor registrant’s filings are attributed to the current ticker. Companies listed later (AbbVie, Meta, PayPal, Uber, Palantir and others) enter when their second 10-K exists.

Filings. 1,792 original 10-K filings for fiscal years 2007 to 2025 (10-K/A amendments excluded), downloaded from the EDGAR submissions index and Archives with the required identifying **User-Agent**, at most eight requests a second and a local cache. FY2007 is used only as the predecessor of FY2008. The primary document is converted to plain text (inline-XBRL hidden blocks and CSS-hidden elements dropped, line breaks only at block elements so that headings are not split). Items 1A and 7 are located by pairing each “Item 1A. Risk Factors” / “Item 7. Management’s Discussion” heading with the next “Item 1B/2” / “Item 7A/8” heading and keeping the longest span; table-of-contents entries and cross-references produce short spans and lose. Extractions shorter than 2,000 characters are treated as missing: Item 7 is missing in 23% of pairs (banks and some industrials incorporate the MD&A by reference to the annual report exhibit) and Item 1A in 8%.

Prices and factors. Adjusted daily closes from yfinance for the 99 tickers and SPY from January 2007; daily Fama-French five factors and momentum from the Kenneth French Data Library, compounded within calendar months.

3 Method

Similarity. For each pair of consecutive 10-Ks of the same company (report dates 270 to 500 days apart), and for the full document, Item 1A and Item 7, two measures are computed on lower-case alphabetic tokens of at least two letters: cosine similarity between TF-IDF vectors and Jaccard similarity between the sets of distinct terms. The inverse document frequency is computed *point in time* from all 10-Ks filed in months before the later filing, so a score never depends on documents that were not yet public. This yields 1,690 pairs (1,560 with Item 1A, 1,297 with Item 7).

Before any portfolio is formed, the first result is descriptive: cosine similarity on the full document is nearly saturated, with a median of 0.996 and an interquartile range of 0.994 to 0.998. Annual reports of the largest US companies are long and mostly boilerplate, so the sort is made on the third decimal. Jaccard has more dispersion (median 0.88, IQR 0.85 to 0.91), and its median rose from 0.84 in FY2008 to 0.91 in FY2025 (Figure 1).

Portfolios. Filings are grouped into fiscal-year cohorts (the calendar year in which the reporting period ends; periods ending in the first two weeks of January belong to the previous year). Within each cohort, companies are sorted on the score into quintiles; Q5 holds the least-changed reports. A stock enters its quintile portfolio at the close of the first trading day of the month after its filing date and is held for twelve months, so the portfolio at any date averages the overlapping annual cohorts. Portfolios are equal-weighted; the long-short portfolio is Q5 minus Q1 and is defined only when both legs hold at least five stocks (about 19 on average, minimum 14). Transaction costs are 10 basis points per unit of one-way turnover on each leg; realised turnover is 3.2 per year for the two legs together.

Ranking within a fiscal-year cohort means that an early filer’s quintile depends on scores of firms filing later in the same season. Prices are never used before the filing date (the test suite

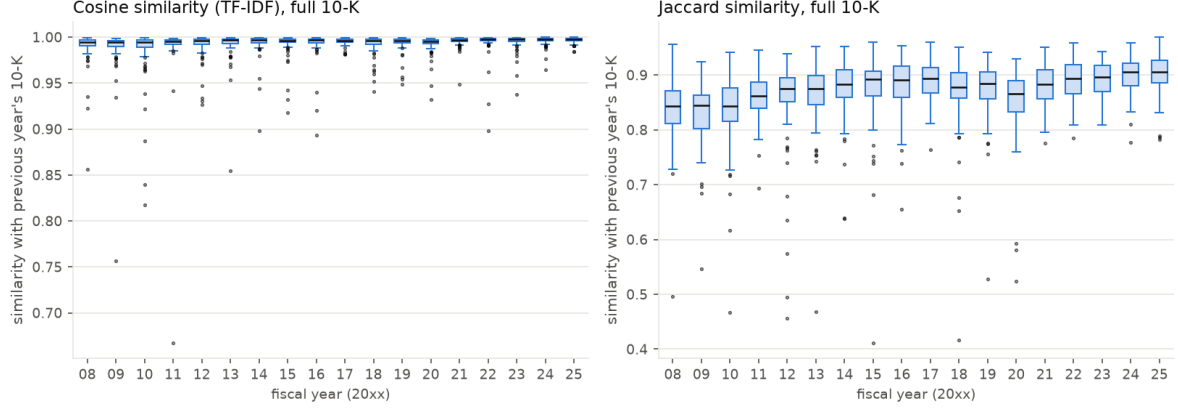


Figure 1: Similarity between consecutive 10-Ks of the same company, by fiscal year.

checks the formation calendar), but the ranking is not strictly real time; a fully point-in-time variant that ranks each filing against the trailing twelve months of filings is reported as a robustness check.

Evaluation. Monthly returns of each quintile (in excess of the risk-free rate) and of the long-short portfolio are regressed on the CAPM, Fama-French three-factor and Fama-French five-factor plus momentum models with Newey-West standard errors (automatic lag $\lfloor 4(T/100)^{2/9} \rfloor = 4$). Sharpe ratios are annualised from monthly returns; a circular block bootstrap (blocks of six months, 2,000 draws) gives their confidence interval. Two model-free checks complement the portfolios: the cumulative market-adjusted return of each quintile by event month after formation, and the within-cohort Spearman correlation between the score and the twelve-month forward return, averaged across cohorts.

Variants and selection bias. The grid of variants is measure (cosine, Jaccard) $\times$ section (full, Item 1A, Item 7) $\times$ groups (terciles, quintiles, deciles) $\times$ formation lag (0, 1, 2 months): 54 trials. Their long-short returns on the common sample (201 months) feed the deflated Sharpe ratio of Bailey and López de Prado (2014), with the effective number of trials estimated from the participation ratio of the trial correlation matrix, and the probability of backtest overfitting via combinatorially symmetric cross-validation (Bailey et al., 2017; $S = 16$, 12,870 partitions). A placebo re-runs the main specification 200 times with scores permuted within cohorts, which preserves every breakpoint and only breaks the link between text and company.

4 Results

4.1 Main specification

The long-short portfolio returns -0.92% a year before costs with a Sharpe ratio of -0.11 (bootstrap 95% interval $[-0.56, 0.38]$) and an FF5+MOM alpha of -0.93% ($t = -0.49$); the CAPM and FF3 alphas are -0.68% ($t = -0.37$) and -0.42% ($t = -0.23$). Costs subtract about 3.2×10 bp a year. SPY over the same months had a Sharpe ratio of 1.02. Figure 2 shows the cumulative return.

Table 1: Long-short portfolio (Q5 minus Q1), main specification: cosine TF-IDF on the full 10-K, quintiles within fiscal-year cohorts, formation the month after filing, twelve-month hold. 211 months, March 2009 to September 2026. Newey-West t -statistics with 4 lags.

	Gross	Net of 10 bp per unit turnover
Mean return (% p.a.)	-0.92	-1.24
Volatility (% p.a.)	8.46	8.49
Sharpe ratio	-0.11	-0.15
Sharpe ratio, 95% bootstrap interval	[-0.56, 0.38]	[-0.58, 0.34]
Maximum drawdown (%)	-33.10	-34.65
CAPM alpha (% p.a.), t -statistic	-0.68 (-0.37)	-1.00 (-0.53)
FF3 alpha (% p.a.), t -statistic	-0.42 (-0.23)	-0.73 (-0.40)
FF5+MOM alpha (% p.a.), t -statistic	-0.93 (-0.49)	-1.25 (-0.65)

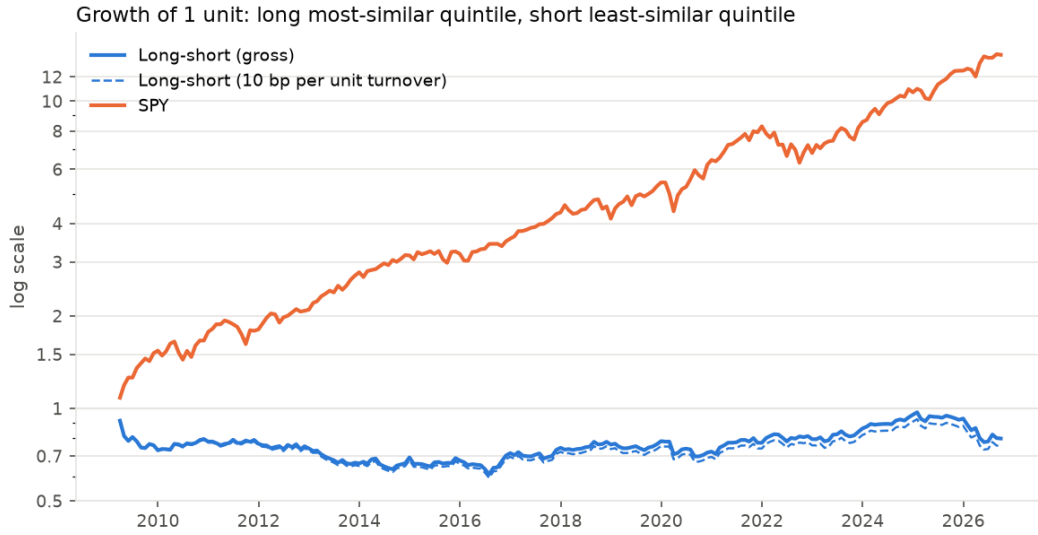


Figure 2: Growth of one unit in the long-short portfolio, gross and net of costs, and in SPY.

Table 2: Quintile portfolios, monthly returns in excess of the risk-free rate. Mean, volatility, drawdown and alphas in percent per year.

Quintile	Mean	Vol	Sharpe	Max DD	CAPM α (t)	FF3 α (t)	FF5+MOM α (t)
Q1 (most change)	19.4	16.5	1.17	-23.0	4.42 (3.00)	4.18 (2.93)	4.54 (3.15)
Q2	22.3	16.3	1.37	-21.5	7.48 (4.24)	7.36 (4.33)	7.73 (4.62)
Q3	16.2	14.6	1.10	-21.4	2.84 (1.89)	2.34 (1.65)	2.20 (1.54)
Q4	19.0	15.3	1.24	-26.4	4.61 (4.34)	4.39 (4.09)	4.60 (4.69)
Q5 (least change)	18.6	16.2	1.15	-28.5	3.66 (2.26)	3.67 (3.31)	3.55 (3.18)

4.2 Quintiles

There is no monotonic pattern across quintiles (Table 2, Figure 3). Every quintile has a positive alpha of 2% to 8% a year. That is not a property of the strategy: it is the survivorship signature of a universe chosen after the fact, and it affects both legs in the same direction.

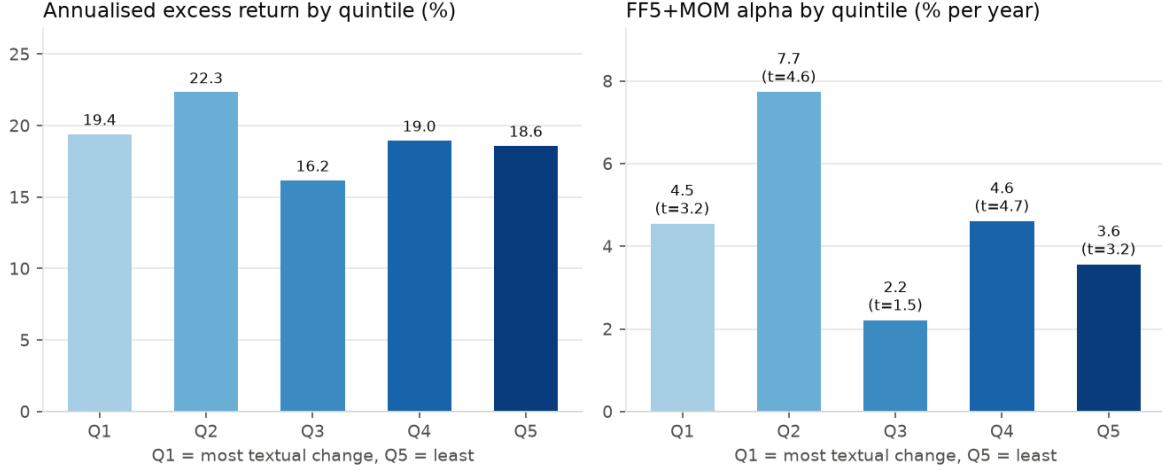


Figure 3: Annualised excess return and FF5+MOM alpha by quintile.

4.3 Event time

Table 3: Cumulative market-adjusted (minus SPY) return in percent by event month after formation, averaged first within and then across fiscal-year cohorts; the standard error treats each cohort as one observation.

Month	Q1 CAR	Q5 CAR	Q5 – Q1	s.e.	Filings
1	0.61	-1.04	-1.65	0.58	1690
3	2.10	-0.85	-2.95	1.39	1690
6	2.19	-0.34	-2.52	1.57	1688
9	2.99	0.30	-2.69	3.09	1614
12	4.18	2.43	-1.75	2.92	1603

The paper’s central figure is the cumulative abnormal return in the months after the filing. Here (Table 3, Figure 4) the least-changed quintile accumulates 2.4% over twelve months against SPY and the most-changed quintile 4.2%; the difference is -1.8% with a t -statistic of -0.60 when each cohort is treated as one observation. The sign is opposite to the paper and the magnitude is inside the noise.

4.4 Robustness and a model-free check

The fully point-in-time ranking, and each half of the sample, give the same answer (Table 4). The rank correlations in Table 5 have the paper’s sign for the full document and for cosine on Item 1A but are three to four hundredths in magnitude with t -statistics near one.

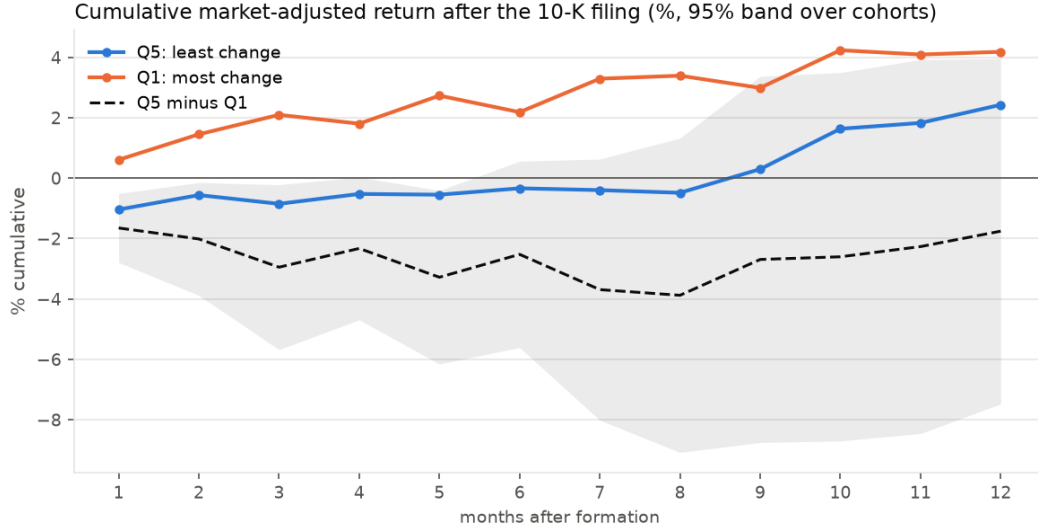


Figure 4: Cumulative market-adjusted return by event month after formation; band is ± 1.96 standard errors over cohorts for the difference.

Table 4: Robustness of the main specification.

Specification	Months	Mean (% p.a.)	Sharpe	FF5+MOM α (t)
main	211	-0.92	-0.11	-0.93 (-0.49)
trailing-window ranking (point-in-time)	214	-1.48	-0.15	-2.73 (-0.87)
main, 2009-2016	94	-3.85	-0.43	-1.27 (-0.52)
main, 2017-2026	117	1.44	0.18	2.09 (1.09)

Table 5: Spearman correlation, within each fiscal-year cohort, between the similarity score and the twelve-month return after formation; time-series mean over cohorts and its t -statistic.

Score	Mean rank correlation	t -statistic	Cohorts positive
Cosine, full document	0.035	1.14	12 of 17
Cosine, Item 1A	0.041	1.00	11 of 17
Cosine, Item 7	0.006	0.18	10 of 17
Jaccard, full document	0.034	0.93	11 of 17
Jaccard, Item 1A	-0.006	-0.15	9 of 17
Jaccard, Item 7	-0.010	-0.30	10 of 17

Table 6: Selection-bias corrections and placebo test.

	Value
Variants tried	54
Variants with a positive Sharpe ratio	17
Variants with $ t > 2$ on the FF5+MOM alpha	8 (all negative)
Best variant on the common sample	<code>cos_1a q10 lag2</code>
Annualised Sharpe, best / median variant	0.54 / -0.05
E[max Sharpe] of N zero-skill trials, raw N	0.80
Deflated Sharpe ratio, raw N	0.13
Effective number of trials (participation ratio)	4.2
E[max Sharpe], effective N	0.37
Deflated Sharpe ratio, effective N	0.76
PBO (CSCV, $S = 16$, 12,870 partitions)	0.08
P[out-of-sample loss of the in-sample winner]	0.21
Placebo: share of 200 shuffles with Sharpe below the actual strategy	0.34
Placebo: standard deviation of the Sharpe ratio across shuffles	0.25
Placebo: share of shuffles with $ t > 2$	0.06

5 Overfitting audit

Of the 54 variants, 17 have a positive Sharpe ratio and 8 have $|t| > 2$ on the FF5+MOM alpha; all 8 of the latter are Item 7 variants with a *negative* alpha, that is, the opposite sign to the paper. The best variant on the common sample is `cos_1a|q10|lag2` with an annualised Sharpe of 0.54 (on its own full sample: alpha 4.30%, $t = 1.34$, maximum drawdown -41%). Reported alone it would look like a modest finding. Against the expected maximum Sharpe of 54 zero-skill trials with the observed dispersion, 0.80, its deflated Sharpe ratio is 0.13; even against the lenient null of 4.2 independent trials (expected maximum 0.37) it is 0.76, below the 0.95 certification threshold (Figure 5).

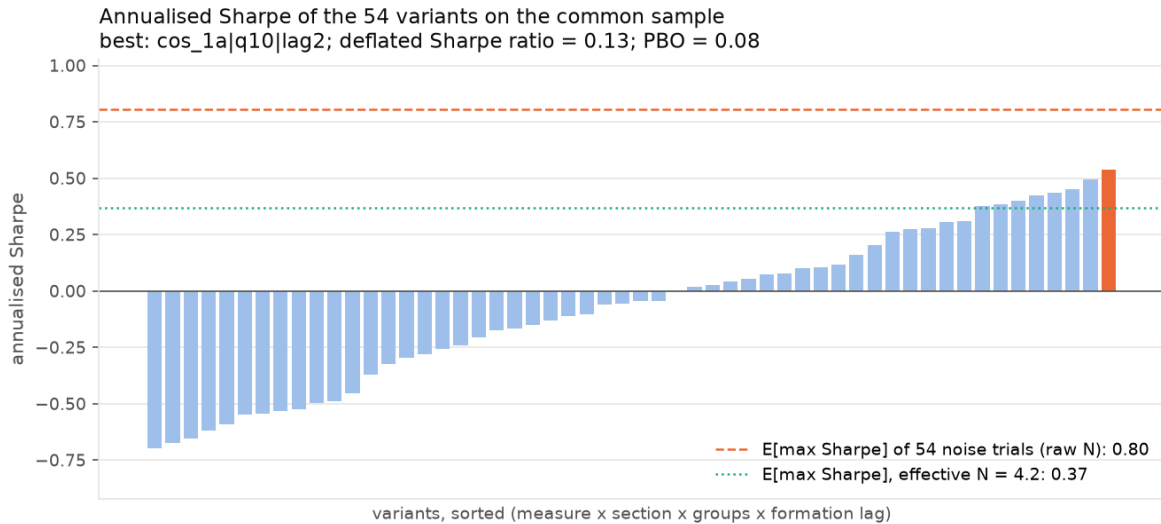


Figure 5: Annualised Sharpe ratio of every variant on the common sample; dashed lines are the expected maximum Sharpe of N zero-skill trials for the raw and effective N .

The probability of backtest overfitting is low, 0.08, and should not be read as reassurance. CSCV asks whether the in-sample winner keeps ranking well *relative to the other variants* out of sample. It does, because the Item 7 variants are consistently the worst and the Item 1A

cosine variants consistently the least bad; relative rankings are stable. PBO measures overfitting within the search, not whether the winner is profitable: the in-sample winner still loses money out of sample in 0.21 of the partitions.

The placebo (Figure 6) shows what the machinery produces when the text carries no information. Shuffling scores within cohorts 200 times gives long-short Sharpe ratios with a standard deviation of 0.25 and a 5th to 95th percentile range of -0.42 to 0.40; the actual strategy sits at the 34th percentile, and 6% of shuffles produce an alpha with $|t| > 2$ on their own, about the nominal size. The best of the 54 variants sits at the 99.5th percentile of this single-specification null, which is exactly why the comparison must be made against the maximum of N trials, as the DSR does, and not against one draw.

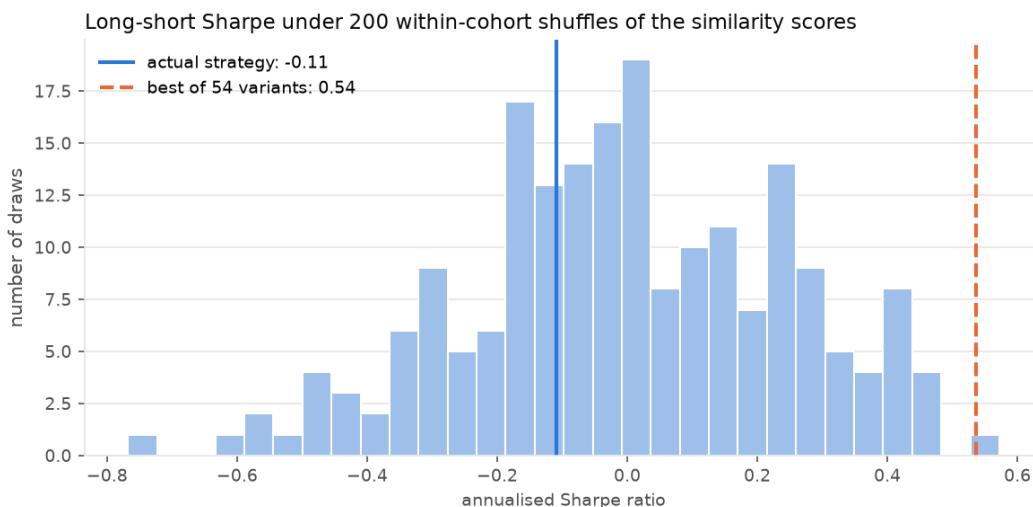


Figure 6: Distribution of the long-short Sharpe ratio when similarity scores are permuted within fiscal-year cohorts.

The worst variant, `jac_7|q5|lag0`, loses -10.0% a year (alpha -9.3%, $t = -2.57$, drawdown -85%). Among these firms the ones that rewrote their MD&A the most did better over the following year. The rank correlations for Item 7 are zero, so this is a tail effect in the extreme groups of a subsample that excludes the banks whose MD&A is incorporated by reference. With 54 correlated trials, 8 of them at $|t| > 2$ in the wrong direction is what dispersion looks like, not evidence of a reverse effect.

6 What I would conclude

1. On the current S&P 100 from March 2009 to September 2026 there is no detectable Lazy Prices effect. The main specification's alpha is slightly negative and indistinguishable from zero under every factor model, with or without costs; the event-time and rank-correlation checks agree.
2. This is not a rejection of the paper. Its effect is estimated on thousands of firms over a period before the idea was published. Here the universe is 100 of the most scrutinised companies in the world, selected after the fact for having survived, in a period after publication. A null in this setting is consistent with the paper's own mechanism: slow attention matters least where attention is greatest.

3. The measurement is weak for this universe. Cosine similarity on the full document is essentially one for every firm every year. Item-level measures help a little but are not always extractable.
4. The best of 54 variants has a Sharpe ratio of 0.54 and a t -statistic above one. The deflated Sharpe ratio (0.13) and the placebo say it is what the luckiest of 54 noise trials would show. Reporting the grid, the DSR and the placebo alongside the winner is the difference between a backtest and a result.

Limitations and next steps. A historical constituent list would remove the survivorship bias; extending the universe to the S&P 500 or to all of EDGAR for a subperiod would give the test power; adding 10-Qs would triple the number of signals as in the paper. None changes the conclusion of this note, which is about what can and cannot be learned from 100 large caps.

References

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A All variants

Table 7: All variants, sorted by annualised Sharpe ratio of the long-short portfolio on each variant’s own sample. Alphas in percent per year (FF5+MOM, Newey-West t).

Variant	Months	Mean (% p.a.)	Sharpe	α	t
cos_full q10 lag2	209	4.19	0.43	3.03	1.34
cos_1a q5 lag2	209	3.62	0.43	3.17	1.70
cos_1a q10 lag2	209	5.25	0.41	4.30	1.34
cos_1a q5 lag1	210	3.40	0.38	3.12	1.60
cos_full q10 lag1	210	3.42	0.33	2.27	0.97
cos_1a q5 lag0	211	2.84	0.31	2.46	1.27
cos_1a q10 lag0	211	4.08	0.31	3.40	1.16
cos_1a q10 lag1	210	3.91	0.30	3.32	1.09
cos_1a q3 lag2	210	1.56	0.22	0.98	0.69
cos_full q10 lag0	211	2.23	0.20	0.89	0.33
cos_1a q3 lag0	212	1.35	0.17	1.01	0.63

Variant	Months	Mean (% p.a.)	Sharpe	α	t
jac_full q3 lag2	212	1.15	0.16	0.84	0.50
cos_full q5 lag2	209	1.04	0.15	0.89	0.58
cos_full q5 lag1	210	0.77	0.10	0.77	0.46
cos_1a q3 lag1	211	0.66	0.08	0.19	0.11
jac_1a q5 lag2	209	0.52	0.06	0.44	0.21
jac_1a q5 lag1	210	0.31	0.04	0.49	0.23
jac_full q5 lag1	210	-0.01	-0.00	0.56	0.29
jac_full q5 lag2	209	-0.03	-0.00	0.10	0.05
jac_1a q5 lag0	211	-0.05	-0.01	-0.01	-0.00
jac_full q3 lag1	213	-0.23	-0.03	-0.53	-0.23
jac_full q10 lag0	209	-0.53	-0.05	1.24	0.56
jac_full q10 lag2	207	-0.51	-0.05	0.28	0.12
jac_full q10 lag1	208	-0.56	-0.05	0.68	0.29
jac_full q5 lag0	211	-0.55	-0.06	0.21	0.10
jac_full q3 lag0	214	-0.69	-0.08	-0.79	-0.34
jac_1a q10 lag1	210	-1.10	-0.10	-0.07	-0.03
cos_full q3 lag2	212	-0.63	-0.10	-1.26	-0.88
cos_full q5 lag0	211	-0.92	-0.11	-0.93	-0.49
jac_1a q10 lag0	211	-1.90	-0.17	-1.07	-0.46
jac_1a q10 lag2	209	-1.92	-0.17	-0.90	-0.37
cos_full q3 lag1	213	-1.59	-0.22	-2.41	-1.10
jac_1a q3 lag2	210	-1.63	-0.24	-1.82	-1.17
cos_7 q3 lag2	210	-1.90	-0.26	-1.58	-0.85
jac_1a q3 lag1	211	-2.11	-0.29	-2.18	-1.25
cos_full q3 lag0	214	-2.25	-0.30	-3.02	-1.27
jac_1a q3 lag0	212	-2.07	-0.31	-2.46	-1.50
cos_7 q3 lag0	212	-2.43	-0.31	-1.34	-0.67
cos_7 q3 lag1	211	-2.39	-0.32	-1.59	-0.85
cos_7 q10 lag2	207	-5.10	-0.34	-3.26	-0.89
cos_7 q10 lag0	209	-6.65	-0.42	-5.24	-1.28
cos_7 q10 lag1	208	-6.79	-0.43	-5.24	-1.40
cos_7 q5 lag0	211	-5.91	-0.54	-4.72	-1.63
cos_7 q5 lag2	209	-5.43	-0.55	-4.11	-1.59
jac_7 q3 lag2	212	-4.89	-0.56	-5.15	-1.99
cos_7 q5 lag1	210	-5.91	-0.56	-4.61	-1.69
jac_7 q10 lag2	209	-8.71	-0.58	-7.94	-2.06
jac_7 q3 lag0	214	-6.04	-0.63	-5.92	-2.13
jac_7 q3 lag1	213	-6.12	-0.65	-6.45	-2.26
jac_7 q5 lag2	212	-7.38	-0.65	-7.06	-2.15
jac_7 q10 lag1	210	-10.42	-0.67	-9.30	-2.35
jac_7 q10 lag0	211	-11.95	-0.75	-10.01	-2.35
jac_7 q5 lag1	213	-9.10	-0.75	-8.91	-2.41
jac_7 q5 lag0	214	-10.02	-0.83	-9.27	-2.57

B Decisions not in the paper

- FY2007 filings are downloaded only to serve as the predecessor of FY2008; one filing per company and fiscal year; amendments ignored.
- IDF weights are point in time. In the first months of the sample this reduces to a plain term-frequency cosine.
- Positions are established at the close of the first trading day of the month after the filing and earn returns from the next day; exit at the close of the first trading day twelve months later.
- The long-short return is undefined when either leg has fewer than five stocks, which removes the last months of 2008.
- Costs are a flat 10 basis points per unit of one-way turnover on each leg; no borrowing cost on the short leg.
- Tokens are alphabetic, so numbers do not count as textual change.